

An experimental transfer matrix method to characterize acoustic materials at high sound pressure levels in airflow environment

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ABSTRACT

A transfer matrix method is proposed for experimental characterization of acoustic materials in the presence of airflow at high sound pressure levels. This method uses two experimental measurements with two different termination loads under flow to derive the transfer matrix coefficients of the tested material. The acoustic pressure and velocity fields upstream and downstream of the material are decomposed into forward and backward traveling waves. Complex models of the wavenumbers are used to account for the damping effect of acoustic waves in the presence of turbulent flow. The components of the transfer matrix of the material are given as function of the pressures and velocities at both faces of the material, which are obtained from the two measurements. The proposed method is validated by comparison with two-source method for different experimental measurements with airflow at high sound pressure levels up to 150 dB and good agreements are obtained. Thus, the proposed method can be used to estimate experimentally the acoustic properties of materials at higher sound pressure excitations in the presence of airflow.

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1. Introduction

The standing wave ratio (SWR) method and the transfer function method using two microphones are two standard measurement methods to access the acoustic properties such as the reflection coefficient and the sound absorption coefficient of a tested material [1,2]. It is well known that the SWR method for a wide frequency range measurement using a linear frequency step is time consuming. The measurement with two-microphone method can be faster than the SWR method and using broadband excitation, it can provide the acoustic properties over a wide frequency range. However, for the accuracy of the measurement results, more than one microphone spacing is needed. Some authors studied the accuracy of the two-microphone method [3–5]. Fujimori et al. [6] presented a least square method based on the data sampled at multiple measurement points. Chu [7] observed that the results of least square method are more accurate than the two-microphone method when the measurement is performed at multiple points. Jang and Ih [8] used multiple microphones method to evaluate the errors induced by the positions of the sensors during the measurement of acoustic properties in duct. Using theoretical and experimental simulations, they observed

smallest error within the effective measurement frequency range when the sensors are placed in equidistant positions. Furthermore, the accuracy of the measurement can be improved and the effective frequency range widened by increasing the number of equidistant sensors. Allam and Åbom [9] proposed a general plane wave decomposition procedure to assess the aeroacoustic characteristics of an open unflanged pipe with flow. They measured the reflection coefficient magnitude and the end correction using six microphones array. It was found that the damping of the plane wave increases strongly in comparison with the damping for a quiescent fluid when the acoustic boundary layer becomes thicker than the viscous sublayer. Holland et al. [10] describe the measurement of the sound power flux in duct with flow using pairs of flush wall mounted microphones. They overcome the signal-to-noise ratio issues due to the high levels of the airflow noise by using swept-sine/selective averaging measurement technique. Munjal and Doige [11] presented a two source-location method for estimating the transfer matrix parameters of an acoustic element in the presence of airflow. They carried out analytical expressions for the two source-location method using four-microphone technique and transfer function approach. They observed that the relations that govern the four-pole parameters of the tested acoustic element for the two source-location method are identical to those of the two-load method [12,13]. However, the two source-location method [11,28] is shown to be more stable and independent of

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the loading terminations [25]. The two-load method described in ASTM E2611-09 [14] required the measurement of the tested element under two different loads that may not be too close to each other at all frequencies of interest. If the two loads are very similar, the result will become prone to error. The use of anechoic termination as one of the two loads will result in minimal reflection of the acoustic wave and the second load can be an open or closed termination. For materials, which are geometrically symmetric, only one load is needed to obtain their acoustic properties [15,16]. Anechoic termination is preferred in this case. The two-load method of ASTM E2611-09 [14] does not consider the effect of airflow and there are no works about this method for the estimation of the transfer matrix and the transmission loss (TL) of an acoustic material in the presence of airflow.

In this paper, a transfer matrix method is presented to characterize experimentally acoustic materials in the presence of airflow and high pressure excitations. Two experimental measurements with two different termination loads under airflow are required in order to determine the transfer matrix coefficients of the tested material. The proposed method is a modification of the existing two-load method of ASTM E2611 [14] to include the effect of airflow. The four components of the transfer matrix of the material are expressed as function of the pressures and velocities at both faces of the material. The description of the present method is presented in Section 2 and the case of grazing incidence is illustrated in Section 3. The validation of the proposed method by comparison with two-source method is shown in Section 4.

2. Description of the proposed method

The proposed method is a transfer matrix approach using two experimental measurements with two different termination loads to estimate the acoustic properties of a tested material in the presence of airflow. It is based on ASTM standard E2611-09 [14], which uses two microphones each on upstream and downstream section of the sample to obtain the components of the transfer matrix through Eqs. (17)–(24). The present method allows the experimental evaluation of the four-pole parameters of an acoustic element in the presence of airflow at high sound pressure levels (SPL). Two different termination loads are required. An anechoic termination is used to cause minimal reflection of the sound wave back down the tube and open termination is created by removing the anechoic termination. The first measurement with airflow which is done with anechoic termination is denoted by load «a» (Fig. 1(a)) and the second measurement which is performed using open termination is denoted by load «b» (Fig. 1(b)). The plane waves inside the tube are decomposed into forward and backward traveling waves A, B, C, D. Four microphones are used for the measurement, two on each side of the material whose thickness is denoted by d . The distance between each pair of microphones is respectively s_1 and s_2 , L_1 is the distance between microphone 2 and the front surface of the sample while L_2 is the distance between microphone 3 and the front surface of the sample. High sound speakers capable of delivering a high sound pressure level up to 150 dB are used on the set up and the airflow through the duct is generated by a compressor.

The two-source method was shown to be more stable and independent of the termination loads [11,25,28] while the two-load method depends mainly on the quality of the two termination loads, which must be different at all the frequencies of interest. However, the two-load method can be preferable for complex acoustic source cases. A motivation for the present proposed method with two loads is to use it for experimental measurements with a harmonic acoustic pneumatic source, which is described in Refs. [32,33]. Fig. 2 shows an experimental setup with the harmonic acoustic pneumatic source to measure the acoustic proper-

ties of materials using four dynamic sensors. Compressed air flows from a plenum chamber through a line, made of a rotating control valve, a flow chopper and the test sample holder section. A compressor generates the air in the plenum and a venturi flowmeter measures the mass flow rate q_m . The alimentation pressure P_{alim} in the plenum is regulated by a regulator. A static pressure sensor, which is positioned before the control valve measures the supply pressure P_o . The flow chopper is a wheel with 6 or 8 apertures, driven in rotation by a brushless motor. It was demonstrated by Allard et al. [32,33] that this acoustic source can generate a very higher sound pressure level up to 180 dB. With this complex acoustic source, it is not possible to use two-source method.

The installation of two harmonic acoustic pneumatic sources in order to use two-source method is not practical therefore; the use of the two-load method with this sound source becomes necessary. This is one of the reasons for developing the present method.

In the following, the methodology to determine the four-pole parameters of the sample (Fig. 1) in the presence of airflow is described and the case where the material is placed at grazing incidence to the airflow and sound waves is discussed further. Using the wave decomposition and omitting the term $e^{j\omega t}$, the acoustic pressure and particle velocity upstream and downstream the sample are expressed as

$$p_u(x) = Ae^{-jk_u^+x} + Be^{+jk_u^-x}, \quad p_d(x) = Ce^{-jk_d^+x} + De^{+jk_d^-x} \quad (1)$$

$$v_u(x) = \frac{1}{Z_u} (Ae^{-jk_u^+x} - Be^{+jk_u^-x}), \quad v_d(x) = \frac{1}{Z_d} (Ce^{-jk_d^+x} - De^{+jk_d^-x}) \quad (2)$$

where j is the imaginary complex number ($j^2 = -1$), Z_u , k_u^+ and k_u^- are respectively the characteristic acoustic impedance and wavenumbers upstream the material, Z_d , k_d^+ and k_d^- the characteristic acoustic impedance and wavenumbers downstream the material.

2.1. Expressions of the wavenumbers upstream and downstream the sample

The propagation of the sound waves in the tube illustrated in Fig. 1 is dissipative because of visco-thermal losses at the walls of the duct. In the present work, the damping of the acoustic wave without airflow is accounted for by modifying the wavenumber $k_o = \omega/c_o$ to $k_o\Gamma_o$ where the propagation constant Γ_o is expressed as [9,17]

$$\Gamma_o = 1 + \frac{(1-j)}{s\sqrt{2}} \left(1 + \frac{\gamma-1}{\sqrt{\text{Pr}}} \right) - \frac{j}{s^2} \left(1 + \frac{\gamma-1}{\sqrt{\text{Pr}}} - \frac{\gamma(\gamma-1)}{2\text{Pr}} \right) \quad (3)$$

with γ the specific heat ratio, Pr is the Prandtl number, $s = R\sqrt{\rho_o\omega/\mu}$ is the shear wavenumber, ω is the angular frequency, μ is the dynamic viscosity, R is the radius of the tube, c_o is the speed of sound in air and ρ_o is the density of the fluid. In the presence of a turbulent flow, the damping of the acoustic waves is affected by the action of the turbulent stresses when the acoustic boundary layer thickness is larger than that the viscous sublayer. The wave numbers $k_u^\pm$ and $k_d^\pm$ upstream and downstream the sample are calculated using the model proposed by Howe [19]

$$k_u^\pm = \frac{\omega/c_o}{(1 \pm M_d)} \left\{ 1 + \frac{(1-j)}{s\sqrt{2}} \text{conj} \left[\frac{1}{(1 \pm M_d)^2} F_A \left(\sqrt{\frac{j\omega\gamma}{k_k^2 u_s^2}}, \delta_v \sqrt{\frac{j\omega}{\gamma}} \right) + \frac{\gamma-1}{\sqrt{\text{Pr}}} F_A \left(\text{Pr} \sqrt{\frac{j\omega\gamma}{k_k^2 u_s^2}}, \delta_v \sqrt{\frac{j\omega}{\gamma}} \right) \right] \right\} \quad (4)$$

$$k_d^\pm = \frac{\omega/c_o}{(1 \pm M_d)} \left\{ 1 + \frac{(1-j)}{s\sqrt{2}} \text{conj} \left[\frac{1}{(1 \pm M_d)^2} F_A \left(\sqrt{\frac{j\omega\gamma}{k_k^2 u_s^2}}, \delta_v \sqrt{\frac{j\omega}{\gamma}} \right) + \frac{\gamma-1}{\sqrt{\text{Pr}}} F_A \left(\text{Pr} \sqrt{\frac{j\omega\gamma}{k_k^2 u_s^2}}, \delta_v \sqrt{\frac{j\omega}{\gamma}} \right) \right] \right\} \quad (5)$$

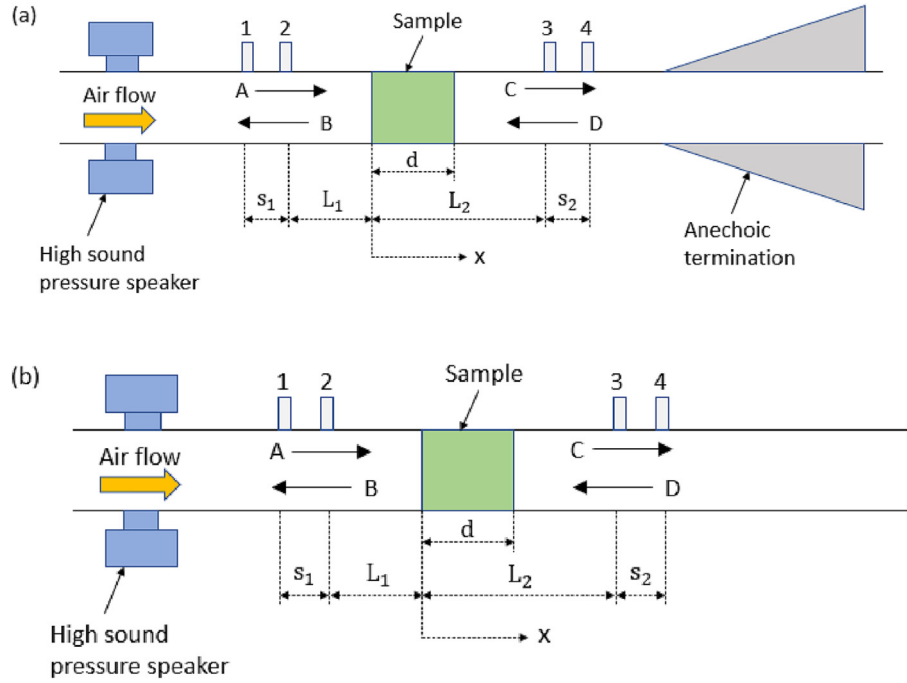


Fig. 1. Sketch of the experimental set up for the measurement with two termination loads: (a) measurement with load «a», (b) measurement with load «b».

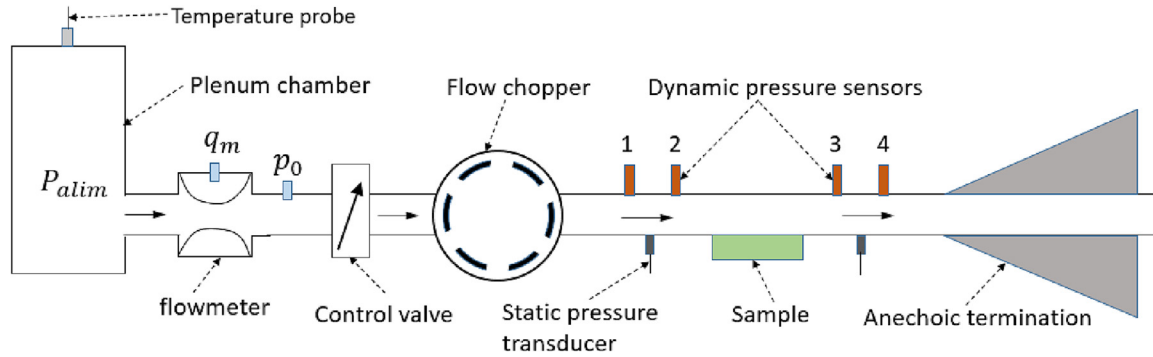


Fig. 2. Experimental set up using harmonic acoustic pneumatic source.

where M_u and M_d are the airflow Mach numbers upstream and downstream the sample, ν is the kinematic viscosity, $\kappa_k \approx 0.41$ is the von Karman constant, «conj» denotes the complex conjugate, $\chi = \kappa_{th}/\rho_0 C_p$ with κ_{th} the thermal conductivity and C_p the specific heat coefficient. The friction velocity u_* is determined from the equation $U_o/u_* = 2.44 \ln(u_* D_p / 2\nu) + 2$ with U_o the velocity of the fluid and D_p the hydraulic diameter, $F_A(a, b) = \frac{j[H_1^{(1)}(a) \cos(b) - H_0^{(1)}(a) \sin(b)]}{H_0^{(1)}(a) \cos(b) + H_1^{(1)}(a) \sin(b)}$ with $H_i^{(1)}$ the Hankel function of the first kind of order i . The viscous sublayer δ_v in Eqs. (4)–(5) is frequency dependent and is calculated from the following relations.

$$\frac{\delta_v u_*}{\nu} = 6.5 \left(1 + \frac{1.7(\omega/\omega_*)^3}{1 + (\omega/\omega_*)^3} \right), \quad \omega_* \nu / u_*^2 \approx 0.01 \text{ and } \omega > 0 \quad (6)$$

The presented work considered the model of Howe [19]. Other models for the damping of waves in ducts can be found in Refs. [18,26–27].

2.2. Determination of the transfer matrix of the tested material

The properties of the air in the tube namely the sound speed, the density and the dynamic viscosity depend on the environmental

conditions such as the temperature and the static pressure. A temperature sensor is used in the tube (Fig. 4) to measure the temperature during the experimental tests. A pressure sensor is used upstream the material and another downstream the sample (Fig. 4) for static pressure measurements during the tests. The speed of sound in air c_o is calculated by [14] $c_o = 20.047\sqrt{273.15 + T}$ where T is the temperature ($^{\circ}\text{C}$). The density of the fluid upstream and downstream the sample are respectively expressed by [14]

$$\begin{aligned} \rho_u &= 1.29 \frac{P_{su}}{101.325} \left(\frac{273.15}{273.15 + T} \right), \\ \rho_d &= 1.29 \frac{P_{sd}}{101.325} \left(\frac{273.15}{273.15 + T} \right) \end{aligned} \quad (7)$$

where P_{su} and P_{sd} are respectively the static pressures (in kPa) upstream and downstream the sample which are measured by the pressure sensors. The kinematic viscosity of the fluid is given as function of the temperature by [20] $\nu = 1.326 \times 10^{-5} [1 + 6.542 \times 10^{-3} T + 8.301 \times 10^{-6} T^2 + 4.84 \times 10^{-9} T^3]$. The air flow Mach numbers M_u and M_d are obtained by the following relations

$$M_u = \frac{Q_m}{\rho_u c_o S}, \quad M_d = \frac{Q_m}{\rho_d c_o S} \quad (8)$$

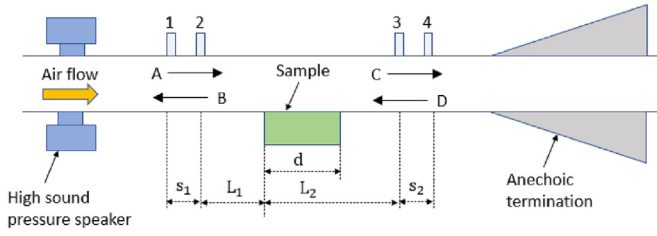


Fig. 3. Grazing incidence experimental set up.

with Q_m the mass airflow rate in Kg/s and S the cross-sectional area of the tube.

High sound pressure speakers are used to generate plane wave field in the standing wave tube. The speakers are capable of delivering a high sound pressure level up to 150 dB. The noise level due to the airflow increases with the flow velocity in the ducts, thus slow sine sweep with a selective averaging technique rather than conventional broadband excitation is used. Four microphones are used to measure simultaneously in the presence of airflow the complex sound pressures P_1 to P_4 at the four measurement locations shown in Fig. 1. The sound pressures at the four measurement locations are deduced from Eq. (1)

$$P_1 = Ae^{-jk_u^+ x_1} + Be^{+jk_u^- x_1} \quad (9a)$$

$$P_2 = Ae^{-jk_u^+ x_2} + Be^{+jk_u^- x_2} \quad (9b)$$

$$P_3 = Ce^{-jk_d^+ x_3} + De^{+jk_d^- x_4} \quad (9c)$$

$$P_4 = Ce^{-jk_d^+ x_4} + De^{+jk_d^- x_4} \quad (9d)$$

where x_1 to x_4 represent the positions of the four measurement locations, they are written as $x_1 = -L_1 - s_1, x_2 = -L_1, x_3 = L_2, x_4 = L_2 + s_2$. The complex pressures P_1 to P_4 are constituted by superposition of forward and backward traveling plane waves in the up- and downstream segments of the standing wave tube. The complex amplitudes of those segments are denoted by the coefficients A to D . From Eqs. (9a) - (9d), one deduces four relations for the coefficients A to D in terms of the four measured sound pressures

$$A = \frac{P_1 e^{+jk_u^- (L_1 + s_1)} - P_2 e^{+jk_u^- L_1}}{e^{+j(L_1 + s_1)(k_u^+ + k_u^-)} - e^{+jL_1(k_u^+ + k_u^-)}} \quad (10a)$$

$$B = \frac{P_1 e^{-jk_u^+ (L_1 + s_1)} - P_2 e^{-jk_u^+ L_1}}{e^{-j(L_1 + s_1)(k_u^+ + k_u^-)} - e^{-jL_1(k_u^+ + k_u^-)}} \quad (10b)$$

$$C = \frac{P_3 e^{-jk_d^- L_2} - P_4 e^{-jk_d^- (L_2 + s_2)}}{e^{-jL_2(k_d^+ + k_d^-)} - e^{-j(L_2 + s_2)(k_d^+ + k_d^-)}} \quad (10c)$$

$$D = \frac{P_3 e^{+jk_d^+ L_2} - P_4 e^{+jk_d^+ (L_2 + s_2)}}{e^{+jL_2(k_d^+ + k_d^-)} - e^{+j(L_2 + s_2)(k_d^+ + k_d^-)}} \quad (10d)$$

For the measurements with the two loads «a» and «b», one defines the transfer function $H_{mn} = P_n/P_m$ so that $H_{mna} = P_{na}/P_{ma}$ and $H_{mnb} = P_{nb}/P_{mb}$. Microphone 3 is used here as reference, so the transfer function H_{3na} is $H_{3na} = P_{na}/P_{3a}$ for the measurement with load «a» and $H_{3nb} = P_{nb}/P_{3b}$ with $n = 1, 2, 4$. The complex amplitudes of the pressures for the measurement with termination load «a» are then given by

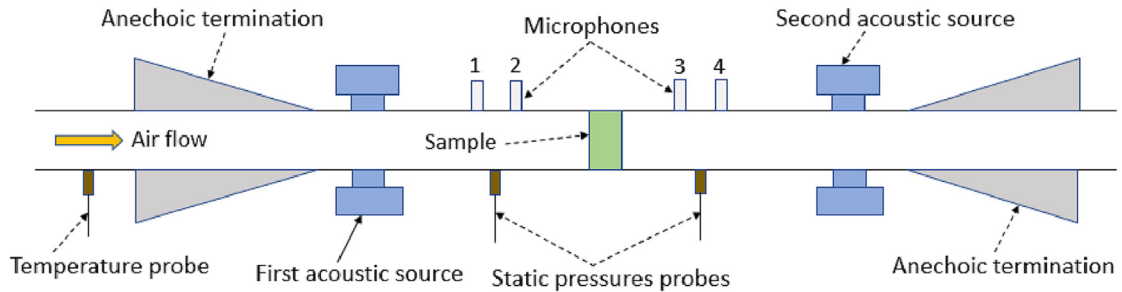


Fig. 4. Sketch of the experimental set up.

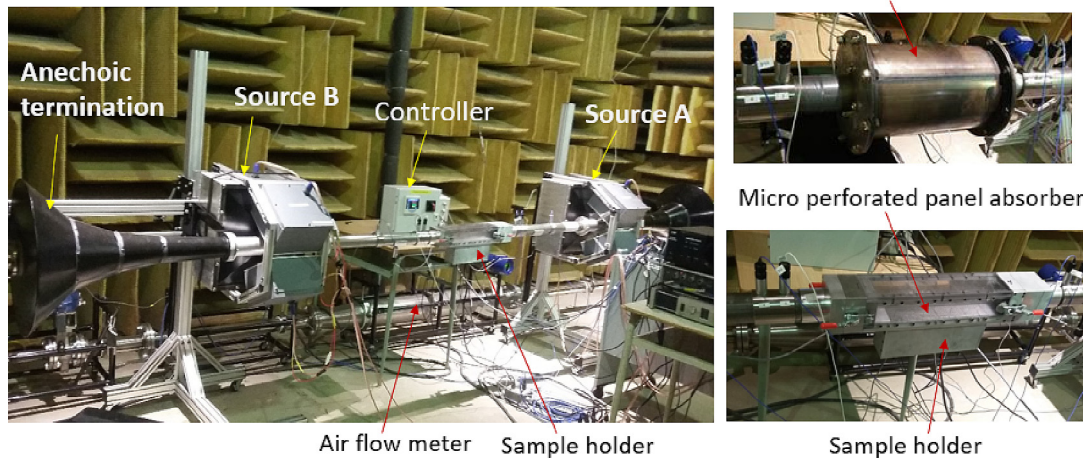


Fig. 5. Experimental set up.

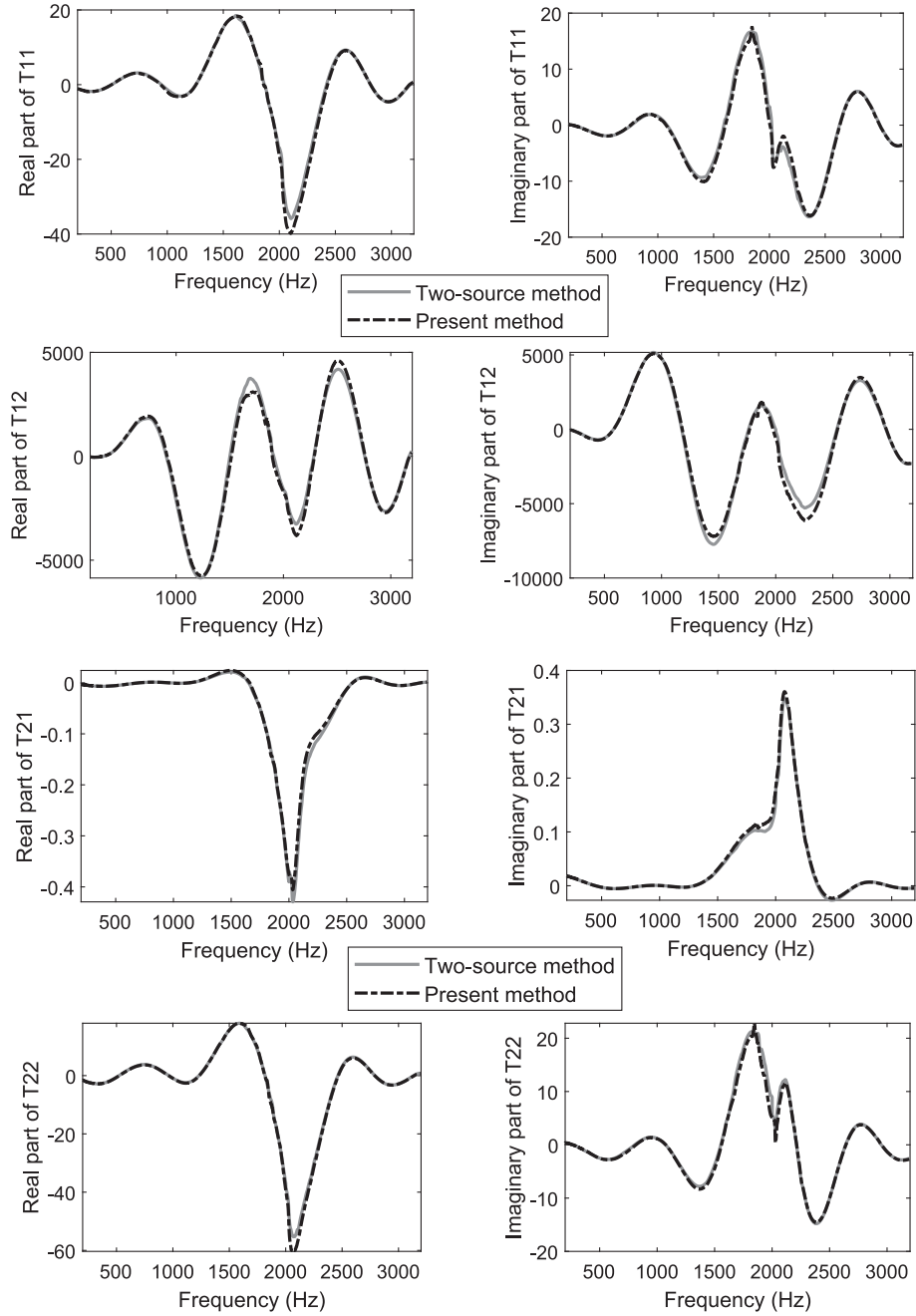


Fig. 6. Transfer matrix components of a perforated silencer filled with fibrous wool. Comparison of the present method with two-source method for $M = 0.015$.

$$A_a = \frac{H_{31a}e^{+jk_u^-(L_1+s_1)} - H_{32a}e^{+jk_u^-L_1}}{e^{+j(L_1+s_1)}(k_u^+ + k_u^-) - e^{+jL_1}(k_u^+ + k_u^-)} \quad (11a)$$

$$B_a = \frac{H_{31a}e^{-jk_u^-(L_1+s_1)} - H_{32a}e^{-jk_u^-L_1}}{e^{-j(L_1+s_1)}(k_u^+ + k_u^-) - e^{-jL_1}(k_u^+ + k_u^-)} \quad (11b)$$

$$C_a = \frac{e^{-jk_d^-L_2} - H_{34a}e^{-jk_d^-(L_2+s_2)}}{e^{-jL_2}(k_d^+ + k_d^-) - e^{-j(L_2+s_2)}(k_d^+ + k_d^-)} \quad (11c)$$

$$D_a = \frac{e^{+jk_d^+L_2} - H_{34a}e^{+jk_d^+(L_2+s_2)}}{e^{+jL_2}(k_d^+ + k_d^-) - e^{+j(L_2+s_2)}(k_d^+ + k_d^-)} \quad (11d)$$

The acoustic pressures and particle velocities on both faces of the sample p_{0a} , u_{0a} at $x = 0$ and p_{da} , u_{da} at $x = d$ from the measurement with load «a» can be determined using Eqs. (11a)-(11d) and Eqs. (1)-(2)

$$p_{0a} = A_a + B_a, \quad p_{da} = C_a e^{-jk_d^+d} + D_a e^{jk_d^-d} \quad (12)$$

$$u_{0a} = \frac{A_a - B_a}{Z_u}, \quad u_{da} = \frac{C_a e^{-jk_d^+d} - D_a e^{jk_d^-d}}{Z_d} \quad (13)$$

For the measurement with termination load «b», the four complex amplitudes of the pressures are written as

$$A_b = \frac{H_{31b}e^{+jk_u^-(L_1+s_1)} - H_{32b}e^{+jk_u^-L_1}}{e^{+j(L_1+s_1)}(k_u^+ + k_u^-) - e^{+jL_1}(k_u^+ + k_u^-)} \quad (14a)$$

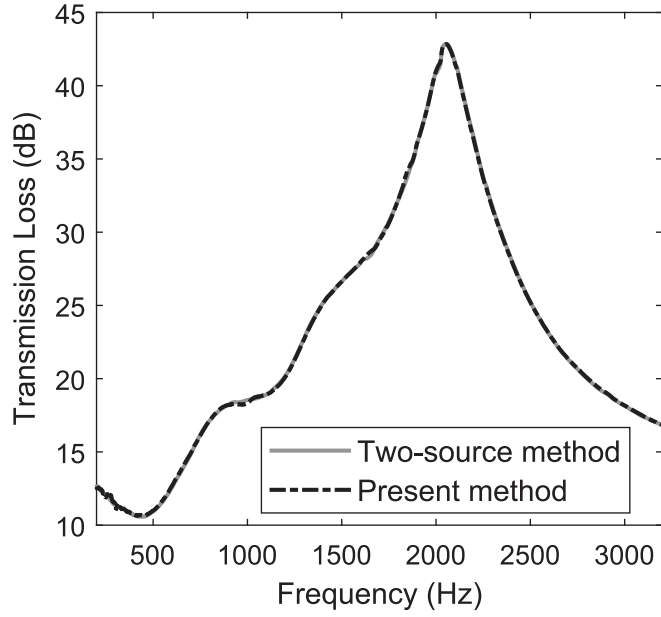


Fig. 7. Transmission loss of a perforated silencer filled with fibrous wool. Comparison of the present method with two-source method for $M = 0.015$.

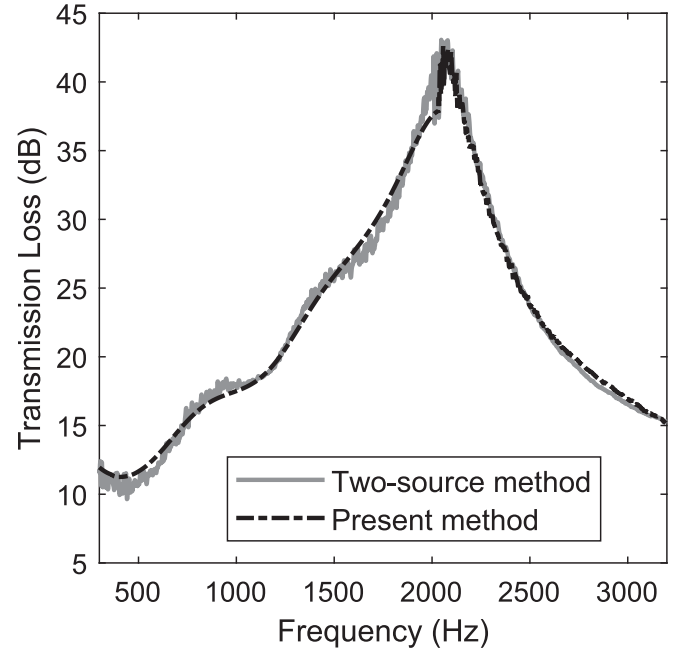


Fig. 9. Transmission loss of a perforated silencer filled with fibrous wool. Comparison of the present method with two-source method for $M = 0.102$.

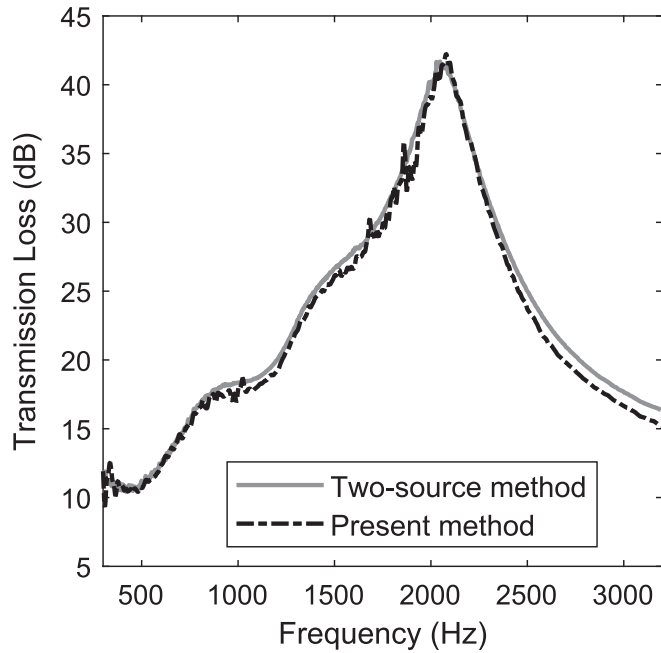


Fig. 8. Transmission loss of a perforated silencer filled with fibrous wool. Comparison of the present method with two-source method for $M = 0.049$.

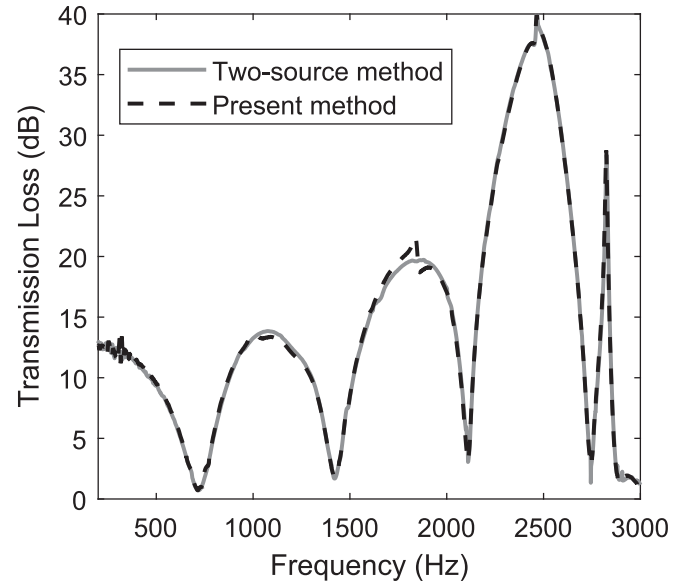


Fig. 10. Transmission loss of a perforated silencer. Comparison of the present method with two-source method for $M = 0.029$.

$$B_b = \frac{H_{31b}e^{-jk_u^+(L_1+s_1)} - H_{32b}e^{-jk_u^+L_1}}{e^{-j(L_1+s_1)(k_u^++k_u^-)} - e^{-jL_1(k_u^++k_u^-)}} \quad (14b)$$

$$C_b = \frac{e^{-jk_d^+L_2} - H_{34b}e^{-jk_d^+(L_2+s_2)}}{e^{-jL_2(k_d^++k_d^-)} - e^{-j(L_2+s_2)(k_d^++k_d^-)}} \quad (14c)$$

$$D_b = \frac{e^{+jk_d^+L_2} - H_{34b}e^{+jk_d^+(L_2+s_2)}}{e^{+jL_2(k_d^++k_d^-)} - e^{+j(L_2+s_2)(k_d^++k_d^-)}} \quad (14d)$$

In the same way, the acoustic pressures and particle velocities from the measurement with load «b» on the two surfaces of the sample (at $x = 0$ and at $x = d$) are given by

$$p_{0b} = A_b + B_b, \quad p_{db} = C_b e^{-jk_d^+d} + D_b e^{jk_d^-d} \quad (15)$$

$$u_{0b} = \frac{A_b - B_b}{Z_u}, \quad u_{db} = \frac{C_b e^{-jk_d^+d} - D_b e^{jk_d^-d}}{Z_d} \quad (16)$$

From the pressures and particle velocities on both faces of the sample in each load case, the transfer matrix of the sample is expressed as

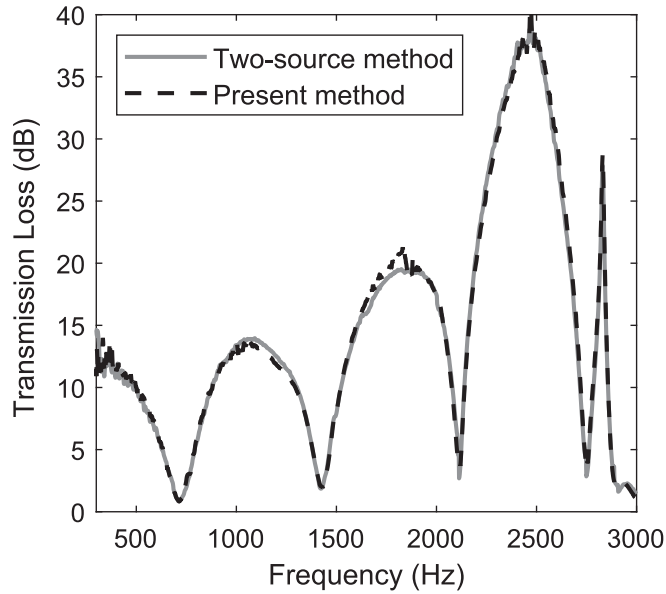


Fig. 11. Transmission loss of a perforated silencer. Comparison of the present method with two-source method for $M = 0.068$.

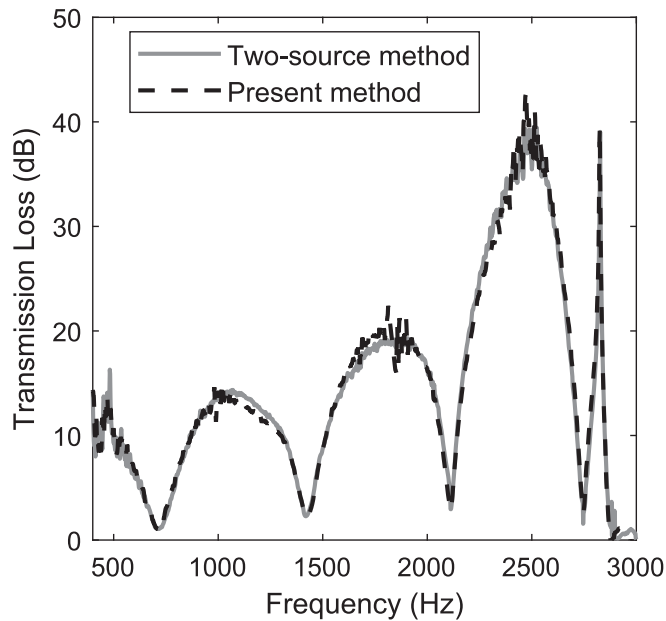


Fig. 12. Transmission loss of a perforated silencer. Comparison of the present method with two-source method for $M = 0.113$.

$$\begin{bmatrix} p_0 \\ u_0 \end{bmatrix}_{a,b} = \begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix} \begin{bmatrix} p_d \\ u_d \end{bmatrix}_{a,b} \quad (17)$$

In Eq. (17), there are four unknown components of the transfer matrix and four equations. By solving this system, the components of the transfer matrix are obtained, they are given by [14,24,34–35]

$$T_{11} = \frac{p_{0a}u_{db} - p_{0b}u_{da}}{p_{da}u_{db} - p_{db}u_{da}} \quad (18)$$

$$T_{12} = \frac{p_{0b}p_{da} - p_{0a}p_{db}}{p_{da}u_{db} - p_{db}u_{da}} \quad (19)$$

$$T_{21} = \frac{u_{0a}u_{db} - u_{0b}u_{da}}{p_{da}u_{db} - p_{db}u_{da}} \quad (20)$$

$$T_{22} = \frac{p_{da}u_{0b} - p_{db}u_{0a}}{p_{da}u_{db} - p_{db}u_{da}} \quad (21)$$

The transmission loss is calculated using the four components of the transfer matrix [24,34–35]

$$TL = 20 \log_{10} \left(\frac{1}{2} \left| T_{11} + \frac{T_{12}}{Z_u} + Z_d T_{21} + T_{22} \right| \right) \quad (22)$$

Note that the expressions of the transfer matrix components in Eqs. 18–21 look similar to the ones of the standard ASTM E2611 but the coefficients A_a, B_a, C_a, D_a for load «a» in Eqs. (11a)–(11d) and A_b, B_b, C_b, D_b for load «b» in Eqs. (14a)–(14d) in the present method account for high SPL and airflow effects. The two acoustic source SPL are controlled experimentally using slow sine sweep with a selective averaging technique. The SPL of each acoustic source in Figs. 4–5 is well controlled. The transfer matrices are measured at each specific SPL since the latter is controlled. The measurements for the loads «a» and «b» use the SPL at the four microphones and thus any non-linear effects are included in the measured transfer functions. These effects are thus included in Eqs. (11a)–(11d) and Eqs. (14a)–(14d). If a material such as a micro-perforated panel absorber is tested, for low POA, the high SPL involves nonlinear behaviors that are taken into account with the measurements of the four microphones. The present experimental method is similar to the standard ASTM E2611 for the measurement of the transfer matrix but the effects of high SPL and airflow are taken into account in the present method. To illustrate this, the effect of the SPL on the transfer matrix is presented in Fig. 20.

3. Characterization at grazing incidence

When the material is placed at grazing incidence to the airflow and to the acoustic waves as shown in Fig. 3, the previous methodology for the determination of the transfer matrix components remains the same. Two measurements are performed with two different termination loads as mentioned previously. The first measurement is done with an anechoic termination that represents the load «a» and the second measurement with the open termination, which is the load «b». The static pressures P_{su} and P_{sd} are measured on both sides of the sample during the experimental test by the pressure sensors installed on either side of the sample (Fig. 4), so the fluid density ρ_u and ρ_d (Eq. (7)), and the Mach air flow numbers M_u and M_d (Eq. (8)) can then be determined as well as the wavenumbers k_u^+ and k_d^+ (Eqs. (4)–(5)). Note that the static pressures P_{su} and P_{sd} are close to each other in this case since the material is placed at grazing incidence, then the wave numbers k_u^+ and k_d^+ will be $k_u^+ \approx k_d^+ = k_z^+$ and $k_u^- \approx k_d^- = k_z^-$, and the air flow Mach number $M_u \approx M_d = M$. Fig. 3 illustrates the measurement set up with load «a». The anechoic termination is removed to create an open termination for the second measurement. The transfer functions $H_{3na} = P_{na}/P_{3a}$ and $H_{3nb} = P_{nb}/P_{3b}$ with $n = 1, 2, 4$ are then obtained from the two measurements and the transfer matrix of the sample is calculated in the same way using Eqs. (18)–(21).

For a measurement at a given sound pressure level L_p , the acoustic source level is well controlled so that the SPL of microphones 1 and 2 corresponds to L_p at each frequency. In all the frequency range, the SPL of microphones 1 and 2 is then L_p .

4. Validation of the proposed method

In the following, the validation of the proposed method by comparison with the two-source method is presented. The four-pole matrix between microphones 2 and 3 are determined in Ref. [11] for the two-source method, they are given by

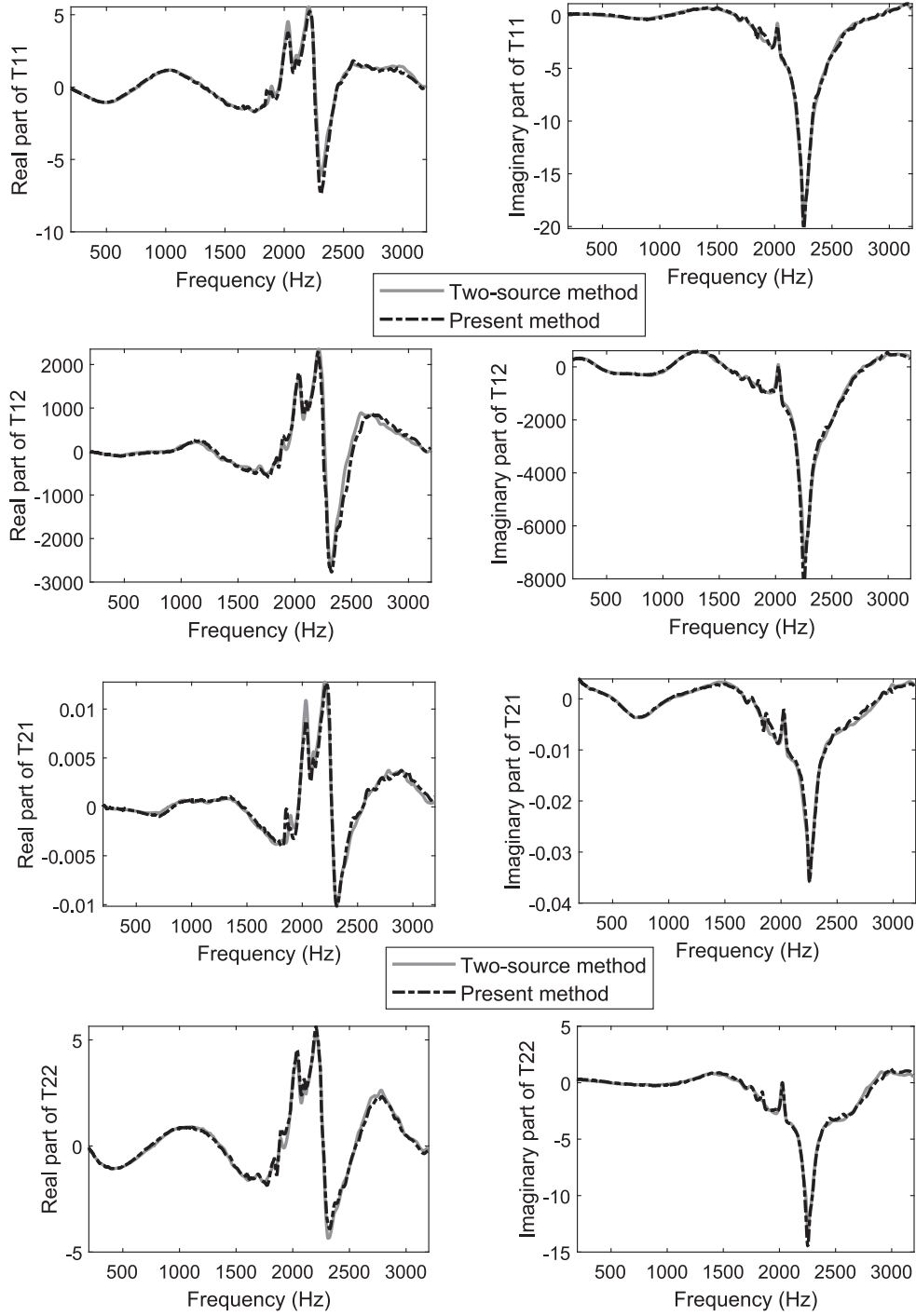


Fig. 13. Transfer matrix components of a perforated panel absorber with slit-shaped holes at grazing incidence. Comparison of the present method with two-source method for $M = 0.0164$.

$$A_{23} = \frac{\Delta_{34}(H_{32a}H_{34b} - H_{32b}H_{34a}) + D_{34}(H_{32b} - H_{32a})}{\Delta_{34}(H_{34b} - H_{34a})} \quad (23)$$

$$B_{23} = \frac{B_{34}(H_{32a} - H_{32b})}{\Delta_{34}(H_{34b} - H_{34a})} \quad (24)$$

$$C_{23} = \frac{(H_{31a} - A_{12}H_{32a})(\Delta_{34}H_{34b} - D_{34}) - (H_{31b} - A_{12}H_{32b})(\Delta_{34}H_{34a} - D_{34})}{B_{12}\Delta_{34}(H_{34b} - H_{34a})} \quad (25)$$

$$D_{23} = \frac{B_{34}(H_{31a} - H_{31b}) + A_{12}B_{34}(H_{32b} - H_{32a})}{B_{12}\Delta_{34}(H_{34b} - H_{34a})} \quad (26)$$

where $A_{ij}, B_{ij}, C_{ij}, D_{ij}$ represent the corresponding four-pole parameters between microphone i and j , $\Delta_{ij} = A_{ij}D_{ij} - B_{ij}C_{ij}$. See Ref. [11] for more details of the two-source method. If TM_u represents the transfer matrix of the section of the tube between the microphone 2 and the sample and TM_d the transfer matrix between the sample and the microphone 3, the transfer matrix TM of the sample using the four-pole parameters of the two-source method is written as

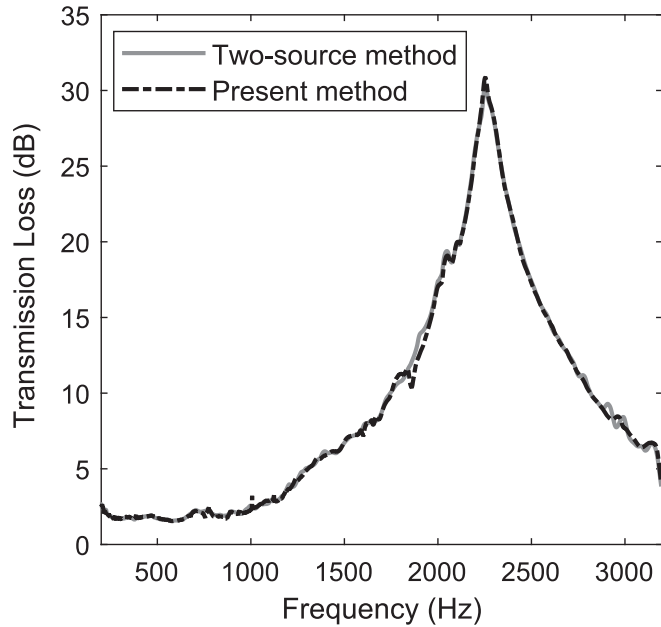


Fig. 14. Transmission loss of perforated panel absorber with slit-shaped holes at grazing incidence. Comparison of the present method with two-source method for $M = 0.0164$.

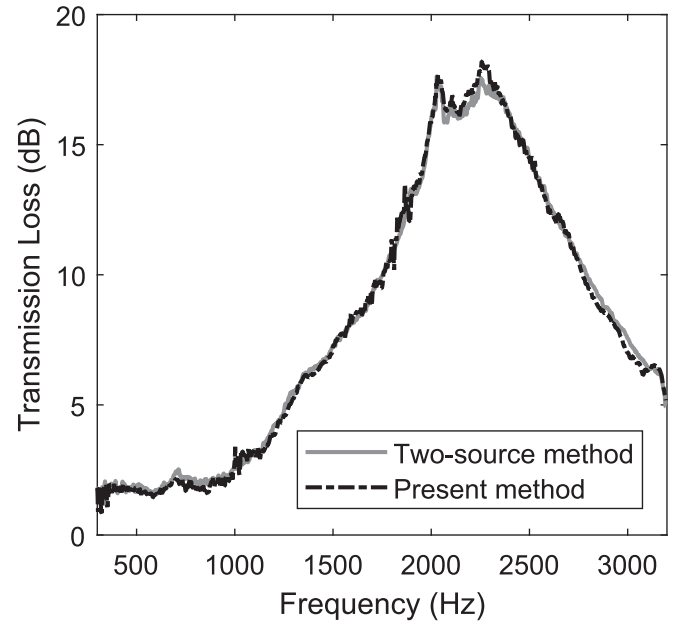


Fig. 16. Transmission loss of perforated panel absorber with slit-shaped holes at grazing incidence. Comparison of the present method with two-source method for $M = 0.103$.

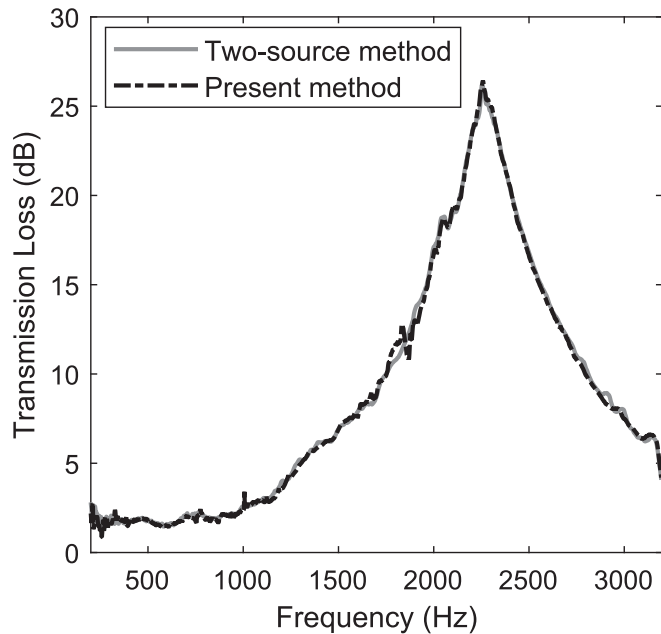


Fig. 15. Transmission loss of perforated panel absorber with slit-shaped holes at grazing incidence. Comparison of the present method with two-source method for $M = 0.053$.

$$TM = (TM_u)^{-1} \begin{bmatrix} A_{23} & B_{23} \\ C_{23} & D_{23} \end{bmatrix} (TM_d)^{-1} \quad (27)$$

The transmission loss is obtained by using the components of TM (Eq. (27)) in Eq. (22). For the validation of the present method, experimental measurements are performed using the experimental set up illustrated in Fig. 4, which is made of two anechoic terminations and two acoustic sources. The first anechoic termination is located before the first source and the second anechoic termination is placed after the second source. Each acoustic source is constituted by two compression drivers BMS 4591 and one loudspeaker BMS 15n620, they can provide a high sound pres-

sure level up to 150 dB. One temperature probe OMEGA HX94C is mounted in the tube before the first anechoic termination and two static pressure sensors MEAS U5244-000005-030PA are installed respectively upstream and downstream of the sample. Four 1/4" microphones PCB 378A14 are mounted flush with the inner wall of the tube, each pair of microphones on one side of the material. A phase and amplitude calibration are done to minimize the error measurement of the microphones. The distance between microphones 1 and 2 is identical to the distance between microphones 3 and 4, which is 40 mm. The inner diameter of the tube is 54 mm. The distance between microphone 2 and the sample is 50 mm as well as the distance between microphone 3 and the sample. The signals from microphones, pressure and temperature sensors and the transfer functions are obtained using a NI CompactDAQ system.

An air compressor generates the airflow in the tube and a pressure regulator adjusts the air pressure. A controller is used to set the airflow rate in the tube.

The measurements with the two loads are performed using the first acoustic source. Anechoic termination is used for the first load and for the measurement with the second load, the anechoic termination is removed to create open termination.

A perforated silencer that contains a single chamber is tested at 145 dB. The inner diameter of the perforated tube is 54 mm, the outer diameter of the silencer is 149 mm, and the length of the perforated tube is 243 mm with a thickness of 1 mm. The diameter of the perforated tube is 3 mm with a perforation ratio of 40%. Fig. 5 shows the perforated silencer and the set up used for the measurements that were carried out in a hemi-anechoic room.

One can see in Fig. 5, the two acoustic sources, the two anechoic terminations, the controller, the airflow meter, the power amplifier and the filter. The sample holder is used to test micro perforated panel (MPP) absorbers. The chamber of the silencer is filled with 394.2 g of fibrous wool with nominal airflow resistivity of $20\,000 \text{ N s m}^{-4}$. Note that the airflow resistivity of the fibrous wool will be affected by its compression ratio inside the silencer. In Fig. 6, the real and imaginary parts of the four matrix components given in Eqs. (18)–(21) are compared with the results of the

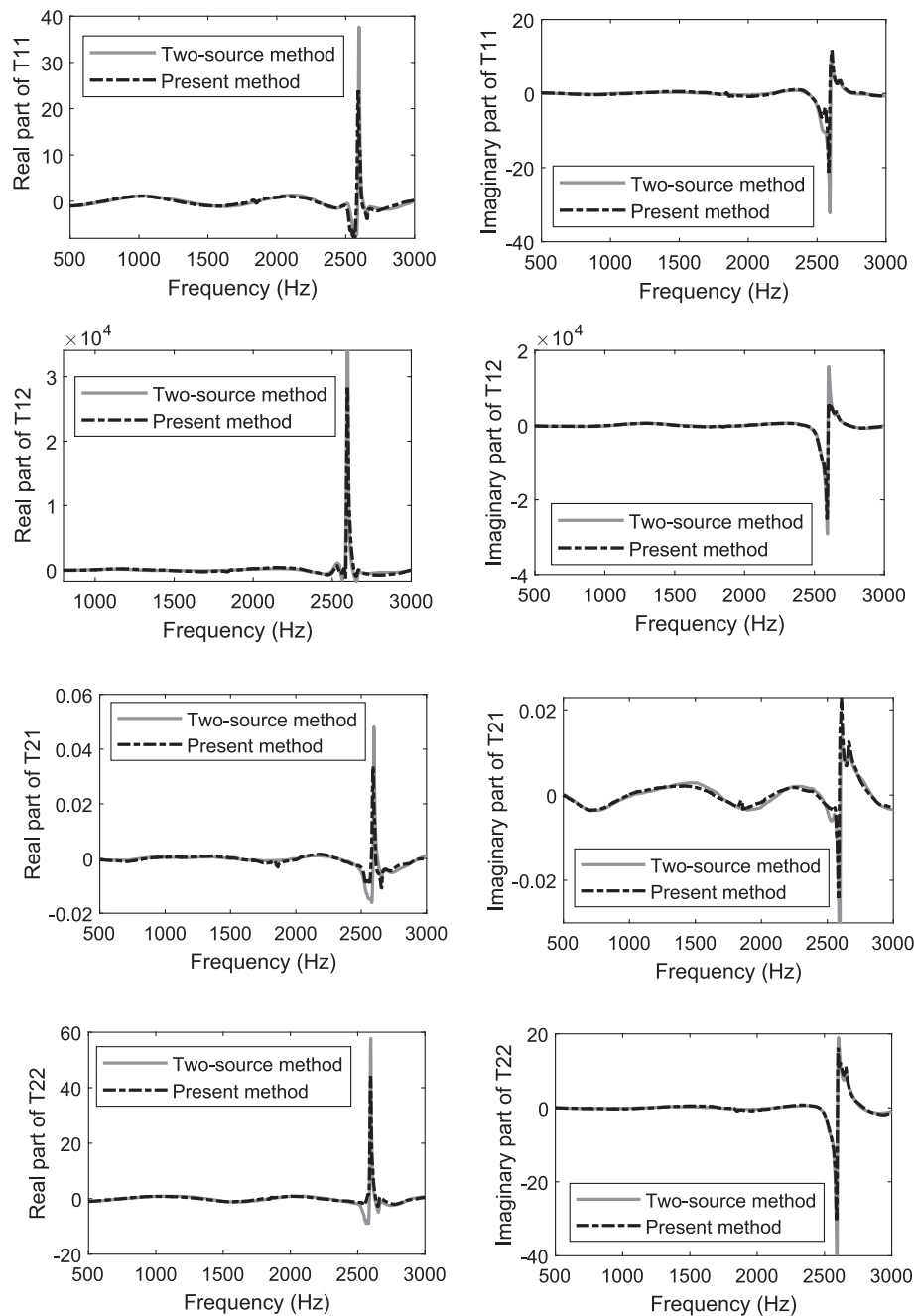


Fig. 17. Transfer matrix components of a perforated panel absorber with circular holes at grazing incidence. Comparison of the present method with two-source method for $M = 0.0248$.

two-source method given in Eq. (27) for mass airflow rate of 48 kg/h, which corresponds to an airflow Mach number of 0.015, the corresponding transmission loss is illustrated in Fig. 7. A good agreement is obtained for all the components of the transfer matrix over the entire frequency range as well as for the transmission loss. The resonant peak of the transmission loss, which is about 43 dB, is observed around 2050 Hz.

Figs. 8–9 show the comparisons of the transmission loss of the previous perforated silencer filled with fibrous wool for respective mass airflow rates of 158 kg/h and 328 kg/h corresponding respectively to airflow Mach numbers of 0.049 and 0.102.

In Figs. 8 and 9, a good correlation between the present method and the two-source method is shown. The peak of the transmission loss remains around the same frequency as the mass airflow rate

increases. In Fig. 7, the TL peak value is 42.8 dB around 2050 Hz for a Mach number of 0.015 while in Fig. 8, the TL peak is 41.3 dB for a Mach number of 0.049. There is therefore a reduction of 1.5 dB for the peak value.

For the next validation case, the previous perforated silencer without the fibrous wool is considered. The chamber of the silencer is therefore empty. The experimental measurements are carried out at 145 dB with mass airflow rates of 93 kg/h, 224 kg/h and 365 kg/h, which give respective Mach numbers of 0.029, 0.068 and 0.113. The comparison of the transmission loss is presented in Figs. 10–12.

Without the fibrous wool, Figs. 10–12 show that the transmission loss of the perforated silencer has different resonant peaks and anti-resonant frequencies which are respectively around

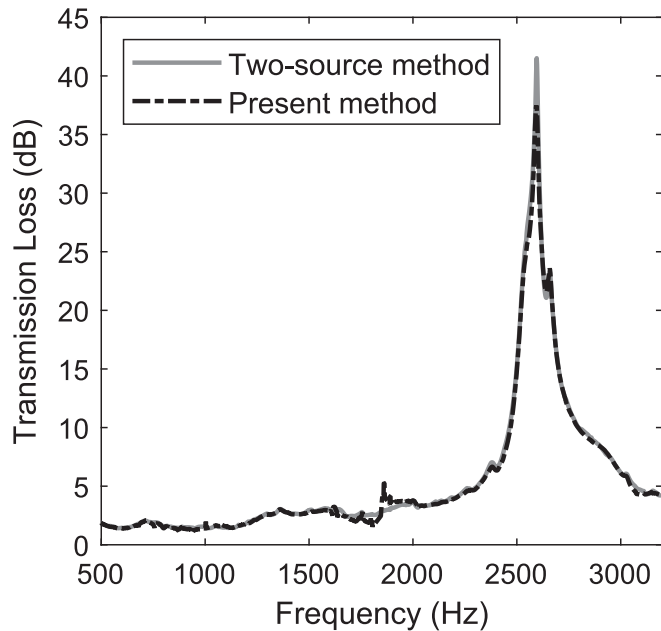


Fig. 18. Transmission loss of a perforated panel absorber with circular holes at grazing incidence. Comparison of the present method with two-source method for $M = 0.0248$.

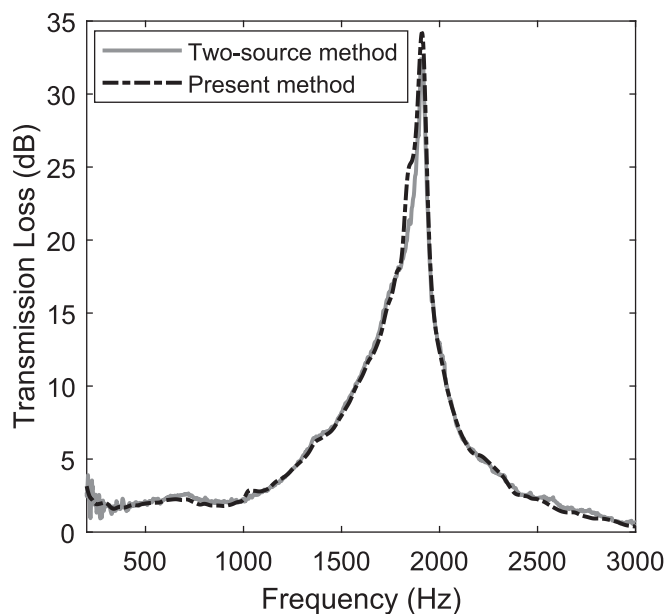


Fig. 19. Transmission loss of a perforated panel absorber with circular holes at grazing incidence. Comparison of the present method with two-source method for $M = 0.102$.

720 Hz, 1424 Hz, 2110 Hz and 2744 Hz where the transmission loss tends towards zero. The present method agrees well with the two-source method. All the transmission loss peaks of the perforated silencer are correctly predicted by the present method. Figs. 9 and 12 illustrate that the quality of the measurements with two terminations loads at higher airflow velocity depends on the quality of the two loads. With a best anechoic termination, the quality of the measurements with the two loads should be better. Note that, the perforation ratio of the silencer tube is 40% with a perforation diameter of 3 mm. At this perforation rate, the behavior of this perforated tube is linear even at the test maximum SPL

around 150 dB. Since the perforation ratio is higher, this will increase the interaction effects between the perforations and reduce the non-linear effects. This may justify the fact that the Figs. 7-9 seem similar, as well as Figs. 10-12.

The following validations are done for grazing incidence cases. The sample is placed at grazing incidence to the airflow and to the acoustic waves as illustrated in Fig. 3. The sample holder has a square cross-section with side dimension of 50.8 mm, it presents a sample location with dimensions of 50.8 mm $\times$ 254 mm. The cavity behind the sample can be adjusted with a plunger. The sample holder is mounted on the experimental set up using two adapters that ensure a gradual transition from the circular section to the square section. The static pressures upstream and downstream of the sample are measured with the two pressure sensors shown in Fig. 4. The measurements with the first load are performed with anechoic termination and the measurements with the second load are done with an open termination as described previously. The first grazing incidence test is carried out on a micro perforated panel absorber. The panel with thickness of 0.6 mm has slit-shaped holes; it is backed by a cavity that is a honeycomb structure with thickness of 17.75 mm and a rigid wall. Fig. 13 shows the real and imaginary parts of the four components of the transfer matrix, the results of the present method are compared with those obtained by two-source method. The measurements are performed at 140 dB with a mass airflow rate of 54 kg/h which is equivalent to an airflow Mach number of 0.0164. The comparison of the corresponding transmission loss is presented in Fig. 14.

The transmission loss of the perforated panel absorber shows a peak of 30.5 dB around 2252 Hz in Fig. 14. At this frequency, one observes also peaks for the coefficients of the transfer matrix in Fig. 13. The present method and the two-source method are in good agreement in Figs. 13 and 14.

Figs. 15-16 show the transmission loss of the previous micro perforated panel absorber with slit shaped holes at 140 dB for mass airflow rates of 170 kg/h and 332 kg/h that correspond respectively to airflow Mach numbers of 0.053 and 0.103. For mass airflow rate of 170 kg/h, the peak of the transmission loss dropped to 26.2 dB in Fig. 15 around the same frequency as before. By increasing the mass airflow rate, the peak of the transmission loss decreases consequently in Fig. 16. Still the present method agrees well with the two-source method.

The last validation case at grazing incidence considers a micro perforated panel absorber with circular holes. The perforation ratio of the panel is 23% with a thickness of 1 mm. The perforated panel is backed by a honeycomb structure with a thickness of 17.75 mm and a rigid wall. The measurements are performed at 140 dB with a mass airflow rate of 80 kg/h, which means a flow Mach number of 0.0248. The comparison of the components of the transfer matrix and the transmission loss are shown in Figs. 17-18. All the real and imaginary parts of the components of the transfer matrix in Fig. 17 show peaks around the frequency of 2600 Hz, which is the resonant frequency of the transmission loss in Fig. 18. Outside the resonant frequency, the components of the transfer matrix are almost zero and the transmission loss with a narrow resonant band tends to zero. The present method shows good agreement with two-source method.

In Fig. 19, the comparison of the transmission loss is presented for the previous perforated panel with circular holes backed by a honeycomb structure with thickness of 31.8 mm and a hard wall. The mass airflow rate for the measurement is set to 328 kg/h, which corresponds to a flow Mach number of 0.102 and the experimental tests are done at 140 dB.

In Fig. 19, the peak of the transmission loss is around 1910 Hz for the micro perforated panel absorber with 31.8 mm thick air backed cavity. In Fig. 18, the peak was observed at 2600 Hz for a cavity with thickness of 17.75 mm. Thus, when the cavity behind

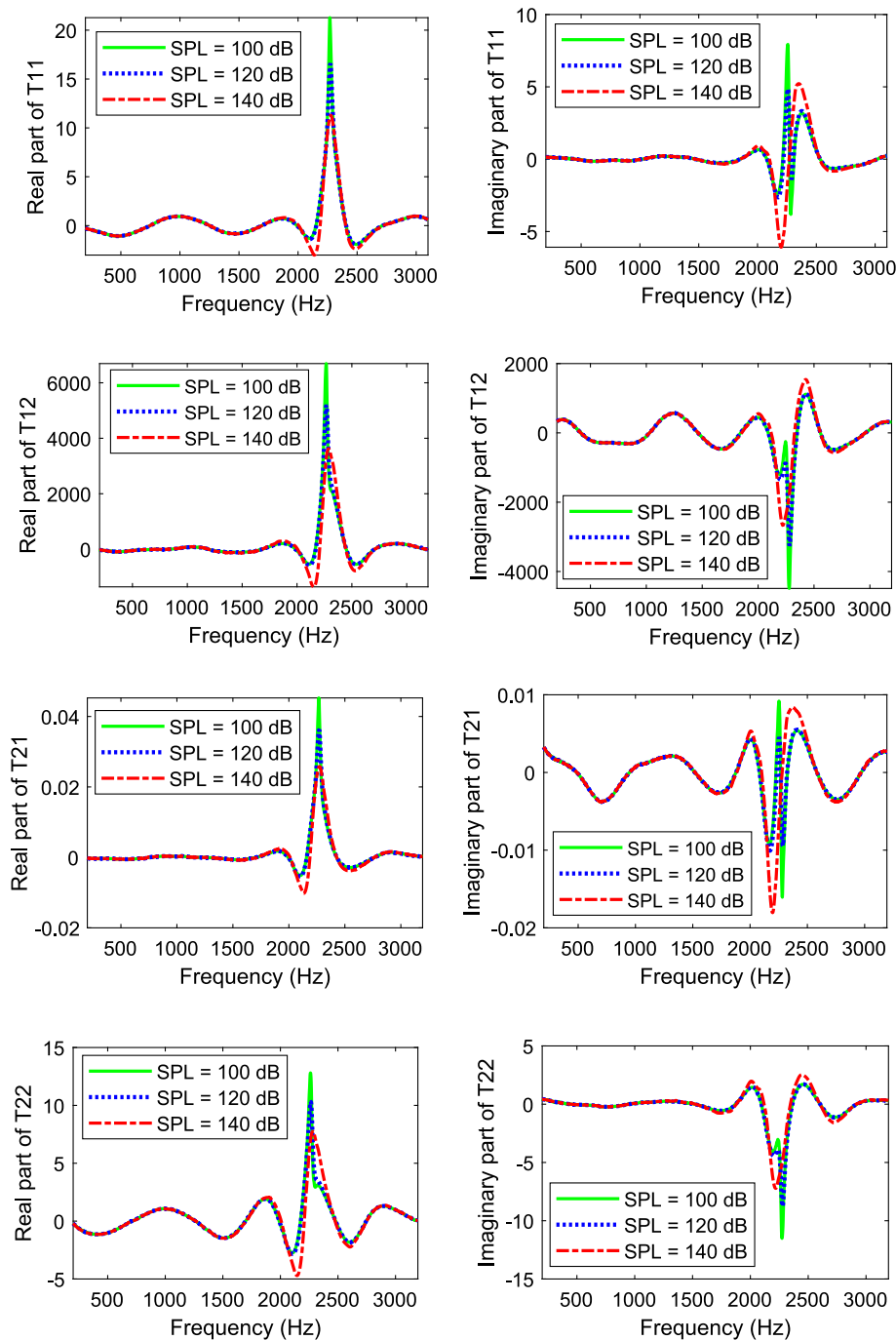


Fig. 20. Transfer matrix components of a micro perforated panel absorber at different SPL.

the micro perforated plate increases, the resonant frequency decreases [21–23]. The present method and the two-source method agree well.

The effect of the sound pressure level on the transfer matrix and the TL of a micro perforated absorber is shown in Figs. 20–21. The micro-perforated panel with a perforation diameter of 1.42 mm, a thickness of 1.3 mm and a perforation open area of 10.4% is backed by a honeycomb structure with cavity depth of 16.6 mm. The MPP absorber is mounted in the sample holder in Fig. 5 and the experimental tests are performed from 100 dB to 140 dB with no flow to evaluate the effects of the SPL on the transfer matrix coefficients and the TL using the present method.

The real parts of T_{11} , T_{12} , T_{21} and T_{22} in Fig. 20 show resonant peaks around 2265 Hz. When the SPL increases, it is observed that the value of the peaks decreases. The TL in Fig. 21 presents a resonance peak around 2265 Hz whose value is 30.8 dB for a SPL of 100 dB. At 120 dB, the TL peak value is 28.7 dB and for a SPL of 140 dB, the peak is 25.8 dB. There is therefore a reduction of 5 dB for the TL peak from 100 dB to 140 dB. This is due to the non-linear phenomena, which appear with the high SPL. Fig. 20 shows that the transfer matrix obtained by the present method depends on the SPL.

High sound pressure levels induce nonlinear phenomena, which affect the acoustic properties of a micro perforated panel.

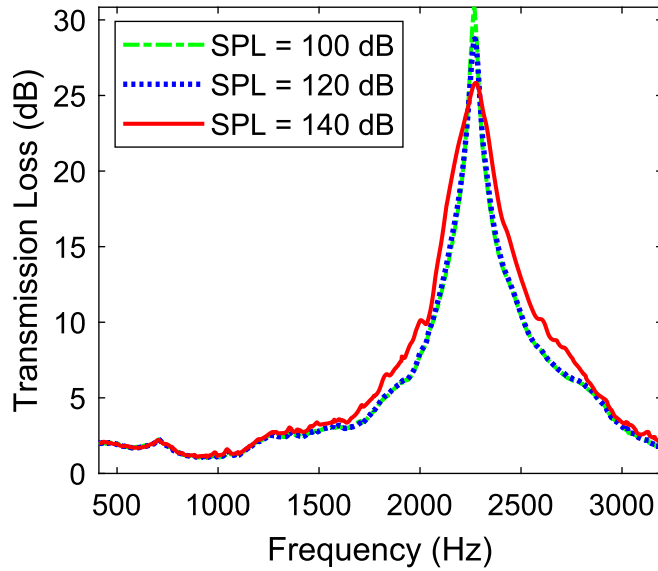


Fig. 21. Transmission loss of a micro perforated panel absorber at different SPL.

A turbulent jet formation at the exit of the orifice was observed by many others [29–30] who showed that the resistance of a MPP at high pressure excitation is a linear function of the particle acoustic velocity in the perforation. The nonlinear phenomena such as vortex and jet formation due to the high SPL increase the acoustic resistance and reduce the orifice end correction and the acoustic energy is converted into kinetic energy of the vortices. The acoustic impedance of a MPP was derived by Laly et al. [21–23] as a function of the incident sound pressure. Using equivalent fluid approach, the normalized acoustic impedance is given by

$$Z_{MPP} = j \frac{\omega t}{\phi c_0} \alpha_{\infty nl} \left(1 + \frac{\sigma_t \phi}{j \omega \alpha_{\infty nl} \rho_0} \sqrt{1 + \frac{4j \rho_0 \omega \mu \alpha_{\infty nl}^2}{\phi^2 \sigma_t^2 \Lambda^2}} \right) \quad (28)$$

with $\alpha_{\infty nl}$ the equivalent tortuosity of MPP, σ_t is the airflow resistivity, ϕ the perforation ratio, Λ the viscous characteristic length that is equal to the radius r of the perforation in the case of cylindrical orifices, t is the thickness of the micro-perforated panel. The flow resistivity σ_t is a function of the incident pressure $\bar{P}_i$ [21–23]

$$\sigma_t = \frac{8\eta}{\phi r^2} + \beta \frac{\rho_0 c_0}{\pi t C_D} \left[-\frac{1}{2} + \sqrt{\frac{1}{4} + \frac{2\sqrt{2}\bar{P}_i (1 - \phi^2)}{\rho_0 c_0^2 \phi^2}} \right] \quad (29)$$

where C_D is the discharge coefficient varying between 0.6 and 0.8 and β a constant. The tortuosity $\alpha_{\infty nl}$ is given by [21–23]

$$\alpha_{\infty nl} = 1 + \frac{2\Psi}{t} 0.48 \sqrt{\pi r^2} \left[\sum_{n=0}^8 a_n (\sqrt{\phi})^n \right] \left[1 + \frac{1}{(1 - \phi^2)} \left(-\frac{1}{2} + \sqrt{\frac{1}{4} + \frac{2\sqrt{2}\bar{P}_i (1 - \phi^2)}{\rho_0 c_0^2 \phi^2}} \right) \right]^{-1} \quad (30)$$

where, Ψ is equal to 4/3 and the parameters a_n are given by

$$a_0 = 1.0, \quad a_1 = -1.4092, \quad a_2 = 0.0; \quad a_3 = 0.33818, \quad a_4 = 0.0 \\ a_5 = 0.06793, \quad a_6 = -0.02287, \quad a_7 = 0.003015, \quad a_8 = -0.01614$$

The normalized surface impedance of the MPP absorber can be obtained by $Z_s = Z_{MPP} - j \cot g(\omega D / c_0)$ with D the cavity depth. In the case of a MPP absorber at grazing incidence in Fig. 3, the trans-

fer matrix of the lined area depends on the sound pressure level since the behavior of the MPP absorber is non-linear. The analytical matrix can be expressed by [31]

$$T_{theo} = \begin{bmatrix} \frac{Z^+ e^{-jk_z^+ L} + Z^- e^{jk_z^- L}}{Z^+ + Z^-} & \frac{Z^+ Z^- (e^{-jk_z^+ L} - e^{jk_z^- L})}{Z^+ + Z^-} \\ \frac{e^{-jk_z^+ L} - e^{jk_z^- L}}{Z^+ + Z^-} & \frac{Z^- e^{-jk_z^+ L} + Z^+ e^{jk_z^- L}}{Z^+ + Z^-} \end{bmatrix} \quad (31)$$

where $Z^\pm = Z_0 \left(\frac{k_0 \mp M k_z^\pm}{k_x^\pm} \right)$, $k_x^\pm = \sqrt{(k_0 \mp M k_z^\pm)^2 - (k_z^\pm)^2}$, $k_0 = \omega / c_0$ and the impedance of the lined wall Z_w verifies this relation [31]

$$Z_w = j Z_0 \frac{k_0}{k_x^+} \left(1 - \frac{M k_z^+}{k_0} \right)^2 \cot g(L k_x^+) \quad (32)$$

With M the airflow Mach number, L_x the lateral dimension of the duct, Z_0 the characteristic impedance of air and L the length of the lined area. For a MPP absorber, the impedance of the lined wall in Eq. (32) is nonlinear and depends on the SPL, therefore the matrix T_{theo} also depends on the SPL. The calculation of this theoretical matrix can be done iteratively using a nonlinear initial impedance given by Eq. (28).

Acoustic properties of materials can be experimentally estimated by the present proposed method that requires two measurements with two different termination loads to determine the four components of the transfer matrix and the transmission loss of the tested materials. The two termination loads should be sufficiently different at all frequencies of interest in order to improve the measurement results. The proposed method can be very useful for the case of a complex acoustic source such as a harmonic acoustic pneumatic source [32–33] to measure the acoustic properties of materials.

5. Conclusion

A transfer matrix method was proposed to estimate experimentally the acoustic properties of sound material in the presence of airflow at higher sound pressure environment. Two experimental measurements with two different termination loads are used to determine the components of the transfer matrix of the tested material. The attenuation of the acoustic waves in turbulent flow conditions is taken into account by the use of complex wavenumber models. The sound pressure level of the acoustic source is well controlled and the nonlinear effects due to the high SPL are accounted for in the transfer matrix measurement. It is shown for a micro perforated panel absorber that the resonant peak values of the real parts of the transfer matrix components decrease as the SPL increases which results in the reduction of the TL peak value. The proposed method is compared experimentally with two-source method at higher pressures excitations with airflow and good agreements are obtained. The present method can therefore predict accurately at high sound pressure levels the acoustic properties of sound materials with airflow. It can be very useful in the case of a complex acoustic source where the two-source method is not practical.

CRediT authorship contribution statement

Zacharie Laly: Conceptualization, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Xukun Feng:** Formal analysis, Investigation, Methodology, Visualization, Writing – review & editing. **Noureddine Atalla:** Supervision, Formal analysis, Funding acquisition, Visualization, Project administration, Resources, Writing – review & editing.

Data availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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